

ĐIỀN ĐẦY ĐỦ THÔNG TIN VÀ MÔ TẢ ĐỀ TÀI ĐỂ SINH VIÊN QUỐC TẾ
CÓ CƠ SỞ LỰA CHỌN INTERNSHIP PROPOSAL

Research Team or Component		The faculty of Water resources engineering
Supervisors	Name/Nom	Vo Ngoc Duong, Doan Thi Ngoc Canh
	Title /Titre	Associated Professor
	Telephone	0905700100
	E-mail	vnduong@dut.udn.vn canhdtm@due.edu.vn ;ngoccanh2710@gmail.com
Workplace		
Internship Category		IUT/Bachelor ☐ Master1/Engineer4 ☒ Master2/Engineer5 ☒
Number of students and specialties (desired pair)		2
Student's name and contact information (if known)		
Proposal title		Salinity forecasting based on AI and Machine Learning

Description of the proposal (2-3 pages max)

1. Introduction:
Salinity intrusion has emerged as one of the most critical challenges affecting coastal and estuarine regions in Central Vietnam, particularly in the downstream areas of the Vu Gia–Thu Bồn River system. This river basin plays a vital role in supporting domestic water supply, agriculture, aquaculture, and industrial development for Quảng Nam Province and Đà Nẵng City. However, in recent decades, the combined effects of climate change, sea-level rise, upstream reservoir operations, and seasonal hydrological variability have intensified the frequency and magnitude of salinity intrusion events. These changes pose significant risks to water security, crop productivity, and socio-economic stability across the region. Traditional hydrodynamic and statistical models have been widely used to simulate and predict salinity distribution. While these methods are effective, they often require extensive datasets, complex calibration, and substantial computational resources. Moreover, their performance may decrease under highly dynamic or unprecedented conditions caused by extreme weather events or operational changes in upstream reservoirs. In this context, Artificial Intelligence (AI) and Machine Learning (ML) techniques offer promising alternatives and complementary approaches. With the ability to learn nonlinear relationships, process large datasets, and adapt to evolving environmental patterns, AI/ML models have demonstrated high potential in forecasting water-related hazards. This study explores the application of AI and Machine Learning for forecasting salinity intrusion levels in the downstream area of the Vu Gia–Thu Bồn River in Vietnam. The objective is to develop data-driven models capable of providing accurate, timely predictions

to support water resources management, irrigation planning, and operational decision-making for hydropower and irrigation reservoirs. By integrating advanced computational methods with local hydrological knowledge, the research aims to contribute to more resilient and adaptive management strategies for the basin under the growing pressures of climate variability and socio-economic development.

2. **Objective:** Application of AI and Machine Learning for Forecasting Salinity Intrusion Levels in the Downstream Area of the Vu Gia–Thu Bon River, Viet Nam
3. **Work to be developed in this internship:** Based on flow data, downstream water levels, and salinity measurements at the Cầu Đỏ station, AI and Machine Learning techniques are applied to forecast salinity levels for the next 12–24 hours
4. **Specific technic tools and working conditions:** AI, ML
5. **Documentation references:**
6. **Appendix (if necessary):**